Introduction

W*indows Internals, Seventh Edition* is intended for advanced computer professionals (developers, security researchers, and system administrators) who want to understand how the core components of the Microsoft Windows 10 and Windows Server 2016 operating systems work internally. With this knowledge, developers can better comprehend the rationale behind design choices when building applications specific to the Windows platform. Such knowledge can also help developers debug complex problems. System administrators can benefit from this information as well, because understanding how the operating system works "under the hood" facilitates an understanding of the performance behavior of the system and makes troubleshooting system problems much easier when things go wrong. Security researchers can figure out how software applications and the operating system can misbehave and be misused, causing undesirable behavior, while also understanding the mitigations and security features modern Windows offers against such scenarios. After reading this book, you should have a better understanding of how Windows works and why it behaves as it does.

History of the book

This is the seventh edition of a book that was originally called *Inside Windows NT* (Microsoft Press, 1992), written by Helen Custer (prior to the initial release of Microsoft Windows NT 3.1). *Inside Windows NT* was the first book ever published about Windows NT and provided key insights into the architecture and design of the system. *Inside Windows NT, Second Edition* (Microsoft Press, 1998) was written by David Solomon. It updated the original book to cover Windows NT 4.0 and had a greatly increased level of technical depth.

Inside Windows 2000, Third Edition (Microsoft Press, 2000) was authored by David Solomon and Mark Russinovich. It added many new topics, such as startup and shutdown, service internals, registry internals, file-system drivers, and networking. It also covered kernel changes in Windows 2000, such as the Windows Driver Model (WDM), Plug and Play, power management, Windows Management Instrumentation (WMI), encryption, the job object, and Terminal Services. *Windows Internals, Fourth Edition* (Microsoft Press, 2004) was the Windows XP and Windows Server 2003 update and added more content focused on helping IT professionals make use of their knowledge of Windows internals, such as using key tools from Windows SysInternals and analyzing crash dumps.

Windows Internals, Fifth Edition (Microsoft Press, 2009) was the update for Windows Vista and Windows Server 2008. It saw Mark Russinovich move on to a full-time job at Microsoft (where he is now the Azure CTO) and the addition of a new co-author, Alex Ionescu. New content included the image loader, user-mode debugging facility, Advanced Local Procedure Call (ALPC), and Hyper-V. The next release, *Windows Internals, Sixth Edition* (Microsoft Press, 2012), was fully updated to address the many kernel changes in Windows 7 and Windows Server 2008 R2, with many new hands-on experiments to reflect changes in the tools as well.

Seventh edition changes

Since this book's last update, Windows has gone through several releases, coming up to Windows 10 and Windows Server 2016. Windows 10 itself, being the current going-forward name for Windows, has had several releases since its initial release to manufacturing (RTM). Each is labeled with a four-digit version number indicating the year and month of release, such as *Windows 10, version 1703*, which was completed in March 2017. This implies that Windows has gone through at least six versions since Windows 7 (at the time of this writing).

Starting with Windows 8, Microsoft began a process of OS convergence, which is beneficial from a development perspective as well as for the

Windows engineering team. Windows 8 and Windows Phone 8 had converged kernels, with modern app convergence arriving in Windows 8.1 and Windows Phone 8.1. The convergence story was complete with Windows 10, which runs on desktops/laptops, servers, XBOX One, phones (Windows Mobile 10), HoloLens, and various Internet of Things (IoT) devices.

With this grand unification completed, the time was right for a new edition of the series, which could now finally catch up with almost half a decade of changes in what will now be a more stable kernel architecture going forward. As such, this latest book covers aspects of Windows from Windows 8 to Windows 10, version 1703. Additionally, this edition welcomes Pavel Yosifovich as its new co-author.

Hands-on experiments

Even without access to the Windows source code, you can glean much about Windows internals from the kernel debugger, tools from SysInternals, and the tools developed specifically for this book. When a tool can be used to expose or demonstrate some aspect of the internal behavior of Windows, the steps for trying the tool yourself are listed in special “EXPERIMENT” sections. These appear throughout the book, and we encourage you to try them as you’re reading. Seeing visible proof of how Windows works internally will make much more of an impression on you than just reading about it will.

Topics not covered

Windows is a large and complex operating system. This book doesn’t cover everything relevant to Windows internals but instead focuses on the base system components. For example, this book doesn’t describe COM+, the Windows distributed object-oriented programming infrastructure, or the Microsoft .NET Framework, the foundation of managed code applications. Because this is an “internals” book and not a user, programming, or system-administration book, it doesn’t describe how to use, program, or configure Windows.

A warning and a caveat

Because this book describes undocumented behavior of the internal architecture and the operation of the Windows operating system (such as internal kernel structures and functions), this content is subject to change between releases.

By “subject to change,” we don’t necessarily mean that details described in this book will change between releases, but you can’t count on them not changing. Any software that uses these undocumented interfaces, or insider knowledge about the operating system, might not work on future releases of Windows. Even worse, software that runs in kernel mode (such as device drivers) and uses these undocumented interfaces might experience a system crash when running on a newer release of Windows, resulting in potential loss of data to users of such software.

In short, you should never use any internal Windows functionality, registry key, behavior, API, or other undocumented detail mentioned in this book during the development of any kind of software designed for end-user systems, or for any other purpose other than research and documentation. Always check with the Microsoft Software Development Network (MSDN) for official documentation on a particular topic first.

Assumptions about you

The book assumes the reader is comfortable with working on Windows at a power-user level, and has a basic understanding of operating system and hardware concepts, such as CPU registers, memory, processes, and threads. Basic understanding of functions, pointers, and similar C programming language constructs is beneficial in some sections.

Organization of this book

The book is divided into two parts (as was the sixth edition), the first of which you’re holding in your hands.

- **Chapter 1**, "**Concepts and tools**," provides a general introduction to Windows internals concepts and introduces the main tools used throughout the book. It's critical to read this chapter first, as it provides the necessary background needed for the rest of the book.

- **Chapter 2**, "**System architecture**," shows the architecture and main components that comprise Windows and discusses them in some depth. Several of these concepts are dealt with in greater detail in subsequent chapters.

- **Chapter 3**, "**Processes and jobs**," details how processes are implemented in Windows and the various ways of manipulating them. Jobs are also discussed as a means for controlling a set of processes and enabling Windows Container support.

- **Chapter 4**, "**Threads**," details how threads are managed, scheduled, and otherwise manipulated in Windows.

- **Chapter 5**, "**Memory management**," shows how the memory manager uses physical and virtual memory, and the various ways that memory can be manipulated and used by processes and drivers alike.

- **Chapter 6**, "**I/O system**," shows how the I/O system in Windows works and integrates with device drivers to provide the mechanisms for working with I/O peripherals.

- **Chapter 7**, "**Security**," details the various security mechanisms built into Windows, including mitigations that are now part of the system to combat exploits.

Conventions

The following conventions are used in this book:

- **Boldface** type is used to indicate text that you type as well as interface items that you are instructed to click or buttons that you are instructed to press.

- *Italic* type is used to indicate new terms.
- Code elements appear in a `monospaced font`.
- The first letters of the names of dialog boxes and dialog box elements are capitalized—for example, the Save As dialog box.
- Keyboard shortcuts are indicated by a plus sign (+) separating the key names. For example, Ctrl+Alt+Delete mean that you press Ctrl, Alt, and Delete keys at the same time.

About the companion content

We have included companion content to enrich your learning experience. The companion content for this book can be downloaded from the following page:

https://aka.ms/winint7ed/downloads

We have also placed the source code for the tools written specifically for this book at **https://github.com/zodiacon/windowsinternals**.

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Errata and book support

We have made every effort to ensure the accuracy of this book and its companion content. Any errors that have been reported since this book was published are listed on our Microsoft Press site at:

https://aka.ms/winint7ed/errata

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